Convergence of Cut Size and Cut Strain on flipped edges of a directed Ring of Cliques Graph

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1 Abstract

In our work, we want to experimentally analyze the convergence of the parameters cut size and cut strain on a directed ring of cliques graph while it is being transformed into an expander by applying a flip chain algorithm. We have chosen the directed ring of cliques graph where each clique is connected to another clique by only one edge to ensure bad connectivity to start with and then apply the flip chain algorithm until the cut size and cut strain of the Graph have converged. At that point the transformation of the original low-conductance graph into an expander has been completed.

2 Introduction

The use of peer-to-peer networks is steadily increasing. Where it was primarily used for file transfer in 1979 (Usenet), it's use has spread to communication transfer like skype VoIP (Salman Baset and Henning Schulzrinne. An analysis of the skype peer-to-peer internet telephony protocol. CoRR, abs/cs/0412017, 2004) or Zattoo (Hyunseok Chang, Sugih Jamin, and Wenjie Wang. Live streaming performance of the zattoo network. In IMC '09: Proceedings of the Ninth ACM SIGCOMM Internet Measurement Conference, pages 417–429, New York, NY, USA, 2009. ACM) and lately also for blockchain technology (Bitcoin: A Peer-to-Peer Electronic Cash System by Satoshi Nakamoto). In peer-to-peer networks you want to provide good connectivity although the nodes in the network are not reliable. Therefor you want to avoid creating bottlenecks where a tiny amount of nodes can create an easy point of failure. A useful graph type for this purpose is an expander graph as it is defined as a sparse graph with good connectivity (wikipedia). The directed flip-chain-operation introduced by Mahlmann and Schindelhauer in 2005 is a tool to transform any d-regular connected graph into an expander. It is defined as series of single flips, where a single flip is redirecting the edges between four chosen nodes. It guarantees to maintain the in- and out- degree of the involved vertices so that a d-regular graph maintains its basic structure throughout the operations. Furthermore, it

is an operation that only includes four adjacent nodes and particularly does not need any further information on the other nodes in the network so it can act locally at all times. Giakkoupis has analyzed flip chains on undirected Graphs in 2022 and obtained an upper bound for the transformation process into an expander in O(nd log2 n) steps, making it an efficient tool to build an expander network. To analyze the efficiency of the flip-chain algorithm, Giakkoupis introduced a new term named "cut strain" that, compared to the cut size, does not only measure the connection between two sets of nodes, but also takes the distribution of the neighbors of each node between the two sets into account. Where the cut size sometimes decreases although the information flow is improving, the cut strain remains a reliable measure. Here, we want to expand the analysis to experiments on directed graphs as peer-to-peer networks consist of nodes with asymmetric connection relations.

3 Experiment

We use a directed ring of cliques graph where n cliques with n nodes each are connected with each other as follows: a single node of a clique removes an outgoing edge to a clique-neighbor (the previously connected node now misses an incoming edge) and creates an outgoing edge to another clique. That selected member of the other clique now has an incoming edge too much to maintain the same number of incoming and outgoing edges, so it removes an incoming edge from one of its own clique-neighbors, that then needs to create an incoming edge to another clique. At the last iteration, the previously connected node from the first clique will receive the created incoming edge from another clique and the ring will be complete as shown in the picture below for a clique size of four.

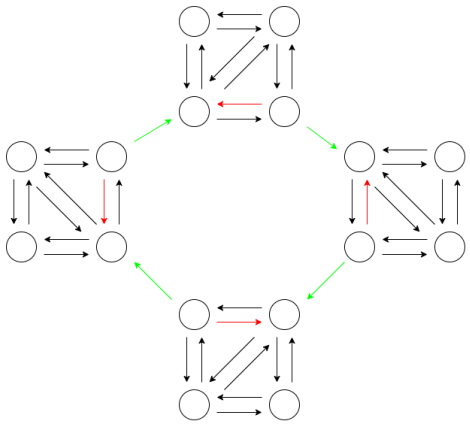


Figure 1: Directed ring of cliques

In comparison to Mahlmann and Schindelhauers definition of a flip-operation, we applied a few changes. The chosen nodes that the flip is performed on are

not allowed to contain a loop and the flip operation is not allowed to create a loop. This has been chosen because the starting graph does not contain loops to start with and creating a loop would mean that a peet is able to send information to itself, which is not practical. It is also ensured, that no edge is created that already exists as double pathing between two peers can change the cut size and cut strain of the graph without increasing the data-relevant connectivity. As consequence, the used algorithm does not perform a random flip, but a valid flip in terms of the constrains that i mentioned above. For our analysis we divide the directed ring of cliques into two sets S and T. The sets are chosen to be as evenly distributed as possible by choosing half of the cliques as S and the other half as T. Should there be an uneven number of cliques, then the set S is receiving one full clique more than the set T. On this graph, we perform 10 000 flip operations and measure the change of cut size and cut strain in each step. For analytical significance, we take 30 equal rings of cliques with the same sets S and T and apply a flip chain of 10 000 steps to each of them separately. The measured cut size and cut strain values are then averaged over the 30 simulations we made.

In the Graph below, you will find the results of experiments with differing clique sizes and number of flip-chain-steps. For very high numbers of flips, we only consider the average parameters of each 30th flip to reduce calculation time and dot density.

4 Results

5 Fazit

References